

BULK CATALYST FOR NITROUS OXIDE DECOMPOSITION IN SPACE THRUSTERS

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Cobalt, manganese, and aluminum oxide-based bulk catalysts for decomposition of nitrous oxide (N_2O) in space thrusters were prepared by the coprecipitation method. Nitrate solutions of cobalt, manganese, and aluminum were coprecipitated in a K_2CO_3 solution at 298 K, with pH adjusted to 10 by adding a KOH solution. The product was filtered, washed, dried, extruded, and calcined at 1173 K. The catalyst performances were evaluated in N_2O decomposition using a 2 N thruster, and the experimental results were compared to theoretical values. In addition, the performances of the bulk catalysts was compared to those of the conventional catalyst (5% Rh_2O_3/Al_2O_3) for N_2O decomposition.

KEY WORDS: mixed oxides, bulk catalysts, N_2O decomposition, thruster

1. INTRODUCTION

Monopropellant thrusters are mainly employed in orbit correction and attitude control of satellites. Most satellites currently in orbit use propulsion systems with hydrazine decomposition and a catalyst composed of 30–40% iridium supported on alumina (Kappenstein et al., 2010). However, hydrazine is a very expensive and highly toxic monopropellant (carcinogenic). In general, satellite propulsion systems employ pressurized gas tanks and solenoid valves to control the monopropellant mass flow rate. On the other hand, there is a class of monopropellants such as nitrous oxide (N_2O) which is saturated at ambient temperature and presents high vapor pressure (~ 5.2 MPa at 293 K). This property gives this monopropellant the capacity of being self-pressurized, avoiding the complexity of a pneumatic or electromechanical feeding system.

It is worth mentioning that N_2O is a high-energy, low-cost, nontoxic propellant, which makes it attractive for application in low-thrust engines in spacecraft (Cai et al., 2012; Zakirov et al., 2001; Kappenstein et al., 2010). Employment of N_2O as a monopropellant is still limited due to the requirement of a catalyst capable of supporting the

extreme conditions during thruster operation and the high temperatures attained in the catalytic bed, which can reach up to 1873 K during N_2O decomposition.

Few studies have been published regarding N_2O catalytic decomposition for space use. Surrey University Space Center reported a series of tests with different catalysts for N_2O decomposition (Baker, 2003; Wallbank et al., 2004; Zakirov, 2000). Among the catalysts tested, the most promising were NiO/ZrO_2 , $\text{IrO}_2/\text{Al}_2\text{O}_3$, and $\text{Rh}_2\text{O}_3/\text{Al}_2\text{O}_3$. All catalysts apparently supported the high temperatures reached in N_2O decomposition (~ 1873 K); however, preheating was necessary to initiate the reaction (~ 773 K). Gaidei et al. (2007) also prepared catalysts for N_2O decomposition from Al_2O_3 impregnation with solutions of H_2PtCl_6 , PdCl_2 , RhCl_3 , RuOHCl_3 , H_2IrCl_6 , $\text{Cu}(\text{NO}_3)_2$, $\text{Co}(\text{NO}_3)_2$, $\text{Ni}(\text{NO}_3)_2$, H_2CrO_4 , $\text{Mn}(\text{NO}_3)_2$, and $\text{Fe}(\text{NO}_3)_3$, varying the content of the active metal between 3 and 30%. Observed ignition temperatures obeyed the following order: Rh (528 K) < Ru (593 K) < Ir (603 K) < Co_3O_4 (688 K) < NiO (698 K) < Pt (703 K) < MnO_2 (703 K) < CuO (713 K) < Cr_2O_3 (718 K) < Fe_2O_3 (738 K) < Pd (763 K). The authors mentioned transition-metal catalysts as a good alternative for preparation of low-cost materials for N_2O decomposition. However, tests above 1073 K were not performed. All prepared catalysts were supported, and thus it can be inferred that during real usage conditions, sintering of the metallic phase can occur and the metal species can be evaporated or carried away.

Zhu et al. (2007) demonstrated that Ir-substituted hexaaluminate ($\text{BaIr}_x\text{Fe}_{1-x}\text{Al}_{11}\text{O}_{19-x}$) is a promising catalyst for N_2O decomposition applicable in propulsion systems. Results showed that the catalyst maintained 100% of N_2O conversion at 773 K for more than 26 h without deactivation. Nevertheless, thruster tests were not performed. All tests were carried out in a fixed-bed flow reaction system using 100 mg of the catalyst and a gas mixture of 30 vol % N_2O in Ar with a flow rate of 50 mL/min.

Russo et al. (2007) recently evaluated the performance of several spinel-type oxides (AB_2O_4) where A = Mg, Ca, Mn, Co, Ni, Cu, Cr, Fe, and Zn and B = Cr, Fe, and Co as catalysts for the N_2O decomposition reaction with environmental purposes. The catalysts were prepared by the combustion method from mixtures of concentrated aqueous solutions of urea and nitrates of the involved species, followed by ignition at 873 K. The oxides containing Co in site B presented higher activity due to its high capacity of forming free sites in the surface, which are responsible for N_2O catalytic decomposition. MgCo_2O_4 catalyst had the best performance and was capable of initiating the reaction at 713 K.

The N_2O decomposition reaction for space purposes is extremely fast, occurring in the vapor phase, in which the thruster ignition speed is very dependent on the number of active sites exposed for the reaction. Since this reaction is extremely exothermic, just a few seconds after ignition, the catalytic bed reaches a temperature high enough so that the reaction ceases to be catalytic and it becomes entirely thermal. The goal of this work is to seek a catalyst with enough active sites to execute an almost instantaneous ignition and to present structural integrity at high temperatures, without significant modifications.

2. EXPERIMENTAL

2.1 Catalyst Preparation

Mixed oxide was prepared from a 1 mol/L metallic ion solution of Co, Mn, and Al in ratio 4:1:1 using nitrates of the cations. Approximately 125 mL of each solution was added, drop by drop, to 50 mL of a 0.5 mol/L potassium carbonate solution, under mixing at ambient temperature. The solution pH was kept at 10 by slow addition of a 3 mol/L KOH solution. After reaction, the reacting medium was mixed for 2 h, followed by a 12 h rest. After aging, the product was washed until it had neutral pH and was vacuum filtered. The material was later dried at 383 K for 12 h and the dry product was grinded to a 100 μm granulometry. This powder was humidified, extruded, dried for 12 h at 333 K, and finally, calcined in air at 1173 K for 1 h.

2.2 Thruster Tests

A thruster prototype (Fig. 1) with theoretical thrust of 2 N was manufactured for operation with N_2O catalytic decomposition. For thruster design, the chamber pressure was set as 500 kPa and the nozzle expansion ratio 1.427 for tests at ambient pressure of 96 kPa, with a throat diameter of 2.1 mm and nozzle exit diameter of 2.55 mm. The calculated theoretical values were adiabatic constant pressure decomposition temperature of 1890 K, characteristic velocity of 1097 m/s, thrust coefficient of 1.1049, and specific impulse of 123.5 s. Catalytic bed dimensions were defined based on 12 mm diameter and 100 mm length for a calculated mass flow rate of 1.65 g/s of N_2O .

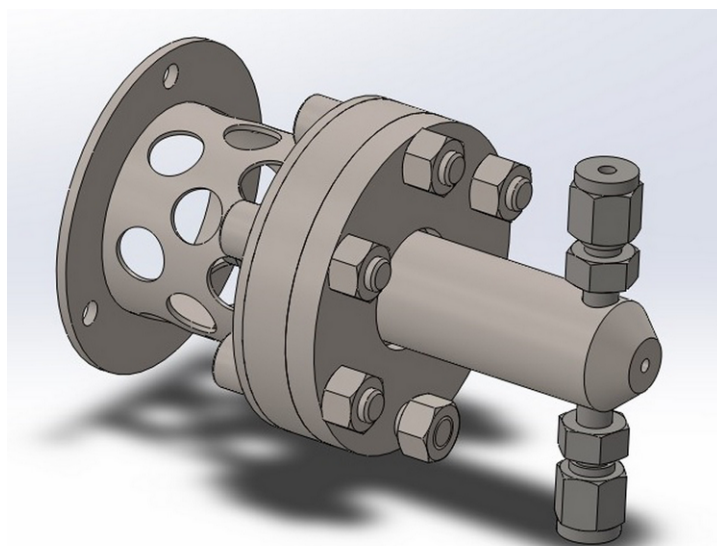


FIG. 1: Computer image of the 2 N thruster prototype for N_2O catalytic decomposition.

3. RESULTS AND DISCUSSION

Mixed oxide was prepared by the coprecipitation method with a molar ratio of Co:Mn:Al (4:1:1) and presented a specific surface of $13 \text{ m}^2/\text{g}$ with an active site density of $31 \text{ } \mu\text{mol}/\text{g}$. The specific surface area and the active site ($\text{Co}^{2+}/\text{Co}^{3+}$) density were determined by N_2 adsorption and CO chemisorption using Micromeritics ASAP2020C.

X-ray diffraction analyses of the catalyst showed the presence of a three-phase mixture of spinel oxides, Co_3O_4 , CoAl_2O_4 , and Co_2MnO_4 , but the Co_3O_4 nonstoichiometric phase was identified as predominant. This is due to the high oxidation potential of the Co^{2+} ion and higher thermodynamic stability of Co_3O_4 in relation to CoO . The chemical analyses of the catalyst surface by XPS also indicates the presence of a Co_3O_4 phase exhibiting Co species on two oxidation levels. Thus it is possible to correlate the oxidation state of the cations present in the surface of the material with its catalytic activity. Therefore it was observed that the presence of a $\text{Co}^{2+}/\text{Co}^{3+}$ redox pair plays an important role in electron transfer, allowing the necessary mobility to increase the activity of the redox surface.

The 2 N thruster was loaded with 7.25 g of the Co:Mn:Al (4:1:1) catalyst. The thruster was lined with an electric resistance to preheat the system at 773 K. Tests were composed of one continuous firing of 10 s and a series of pulsed firings of 5 s.

Figure 2 presents thrust and temperature versus time curves for continuous firing, whereas Fig. 3 depicts the thrust curve for pulsed firings, both cases using the Co:Mn:Al (4:1:1) catalyst. The average thrust ($\sim 1.8 \text{ N}$) for continuous firing was close to the theoretical value of 2 N, whereas the average thrust for pulsed firing was slightly lower ($\sim 1.7 \text{ N}$). Temperatures measured in the tests were around 1473 K (Fig. 2), inferior to the calculated adiabatic reaction temperature ($\sim 1873 \text{ K}$), probably due to thermal losses and incomplete N_2O decomposition. Pulsed tests (Fig. 3) were reproducible.

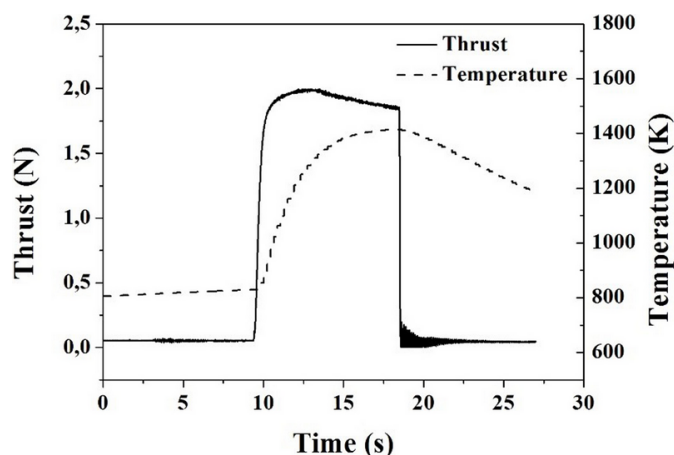


FIG. 2: Thrust and temperature curves for Co:Mn:Al (4:1:1) catalyst tested in continuous mode.

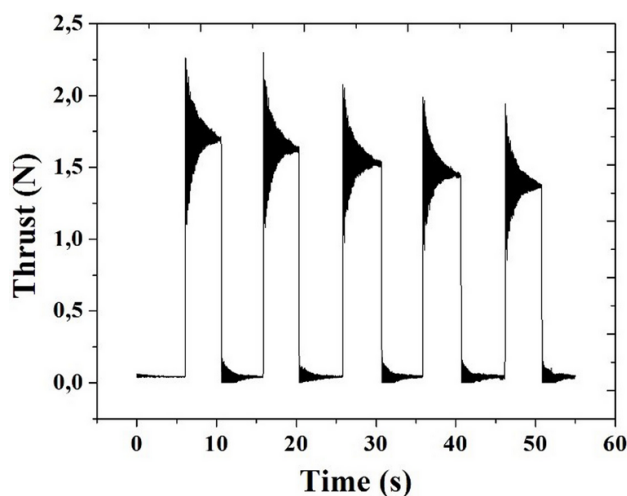


FIG. 3: Thrust curve for Co:Mn:Al (4:1:1) catalyst tested in pulse mode.

A small deactivation of the catalyst was also observed during the tests, possibly due to material contraction and consecutive loss of specific surface, as well as changes in less active phases at the catalyst surface. These phenomena can be caused by high temperatures applied to the catalytic bed. The instability seen in Fig. 3 is apparently due to the thrust balance and data acquisition system.

The characteristic velocity is a parameter in propulsion that evaluates the efficiency of the catalytic system; roughly speaking, it indicates the level of the propellant decomposition. The theoretical value for characteristic velocity is 1096.9 m/s, whereas the experimental value was approximately 945 m/s. It is known that the ideal value for the experimental characteristic velocity must be around 92–99.5% of the theoretical value (Sutton and Biblarz, 2000). In this study, an efficiency for the characteristic velocity of around 86% was obtained, showing that the catalytic decomposition was not complete.

The system is still limited by the high temperatures reached in the reaction. Even though bulk oxides were chosen as catalysts for this thruster system, it was observed that these materials suffer the action of high temperature in their structure. The Co:Mn:Al (4:1:1) catalyst presented a density of 5.12 kg/m³ and a specific surface of 13 m²/g before the tests in the thruster. After 2 min of continuous firing tests the same material presented a density of 5.22 kg/m³ and a specific surface smaller than 1 m²/g. A mass loss of 6% was also detected after the tests.

If bulk catalysts do not support the high temperatures reached by the system, the traditional supported catalysts would not tolerate it either. A sinterization would probably occur and an evaporation of the active metallic phase in these types of catalysts, as well as a contraction and loss of specific area in the catalytic support. Nevertheless, we performed additional tests with catalysts traditionally employed in this reaction, 5% Rh₂O₃/Al₂O₃, and the results were compared with the catalysts developed in this work.

Figures 4 and 5 present thrust curves of the 5% $\text{Rh}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalytic system. The experimental thrust value reached only 2 N at the beginning of the continuous test (Fig. 4), sustaining 1.2 N along the sequence of the experiments. A fast catalyst deactivation was observed as a function of time. The system temperature reached during the tests (~ 973 K) was very inferior to the reaction adiabatic temperature (~ 1873 K).

The experimental characteristic velocity value for this supported catalyst was 785 m/s. Thus an efficiency of around 71% was attained. This value is well below the value obtained with the bulk catalyst (86%), showing a smaller efficiency for the traditional catalytic system. It is also worth mentioning that the catalytic system presented a density of 3.20 kg/m^3 and a specific surface of $185 \text{ m}^2/\text{g}$ before the test. After the test the same

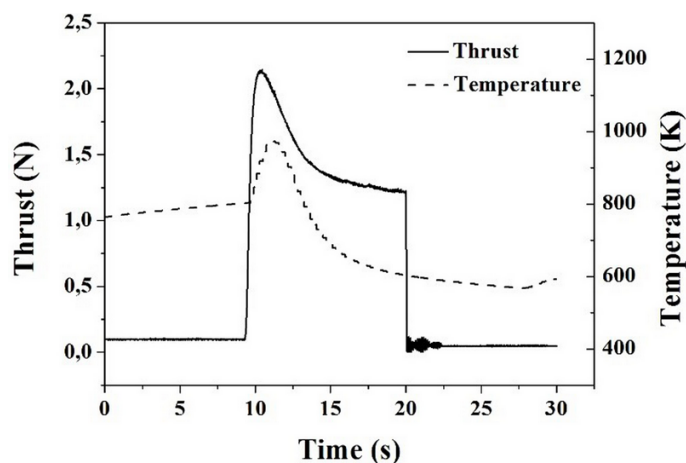


FIG. 4: Thrust and temperature curves for 5% $\text{Rh}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst tested in continuous mode.

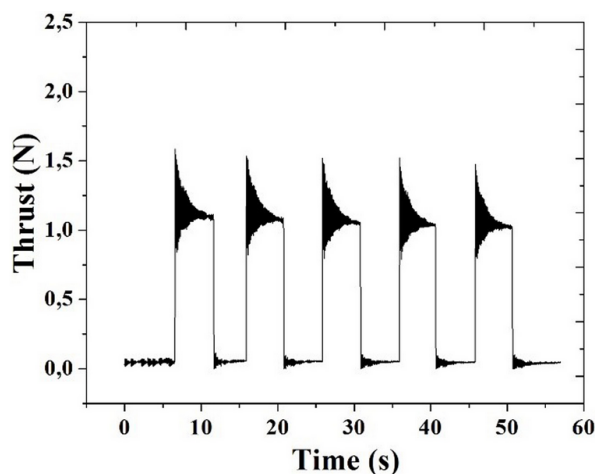


FIG. 5: Thrust curve for 5% $\text{Rh}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst tested in pulse mode.

material presented a density of 3.35 kg/m^3 and a specific surface of $130 \text{ m}^2/\text{g}$, exhibiting a more significant contraction than the one which occurred with the bulk catalyst. A 6% mass loss was also detected after the tests.

4. CONCLUSIONS

Mixed oxide catalysts containing Co, Mn, and Al are easily shaped and very active in the N_2O decomposition. This activity was attributable to the presence of a $\text{Co}^{2+}/\text{Co}^{3+}$ redox pair that facilitates electron transfer. The Co:Mn:Al (4:1:1) catalyst presented better performance in the 2 N thruster than the 5% $\text{Rh}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst traditionally employed in this reaction. Experimental thrust levels were lower than the theoretical value, probably due to thermal losses and incomplete decomposition. Even though the mixed oxide-based catalyst better supported high temperatures, it suffered significant structural modifications during the tests. Despite this and the elevated temperature required for reaction induction, the present bulk catalyst could be applied to hybrid rockets or other thrusters in short-duration missions.

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